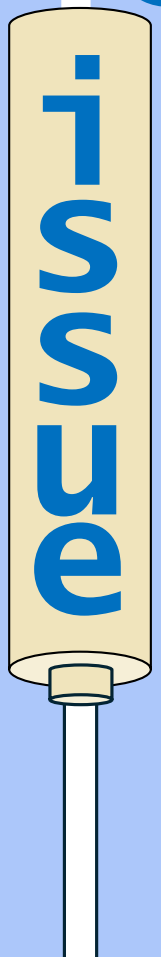
BioTechnology



B2
(BioTwo)

Precision Nutrient Delivery for Tissue Engineering Bioreactors

June 6th 2025

BioT

PRECISION NUTRIENT DELIVERY FOR TISSUE ENGINEERING BIOREACTORS

EMS690U – INTEGRATED DESIGN PROJECT

Presented by: BioT

Group Members:

[redacted]

Yemesis Narkor Addo

[redacted]

June 6th 2025



Project Brief

Problem

High costs and inconsistent performance of current bioreactor systems hinder effective treatment and delay large-scale adoption

Mission

Improve reliability, adaptability and scalability of bioreactors

Solution

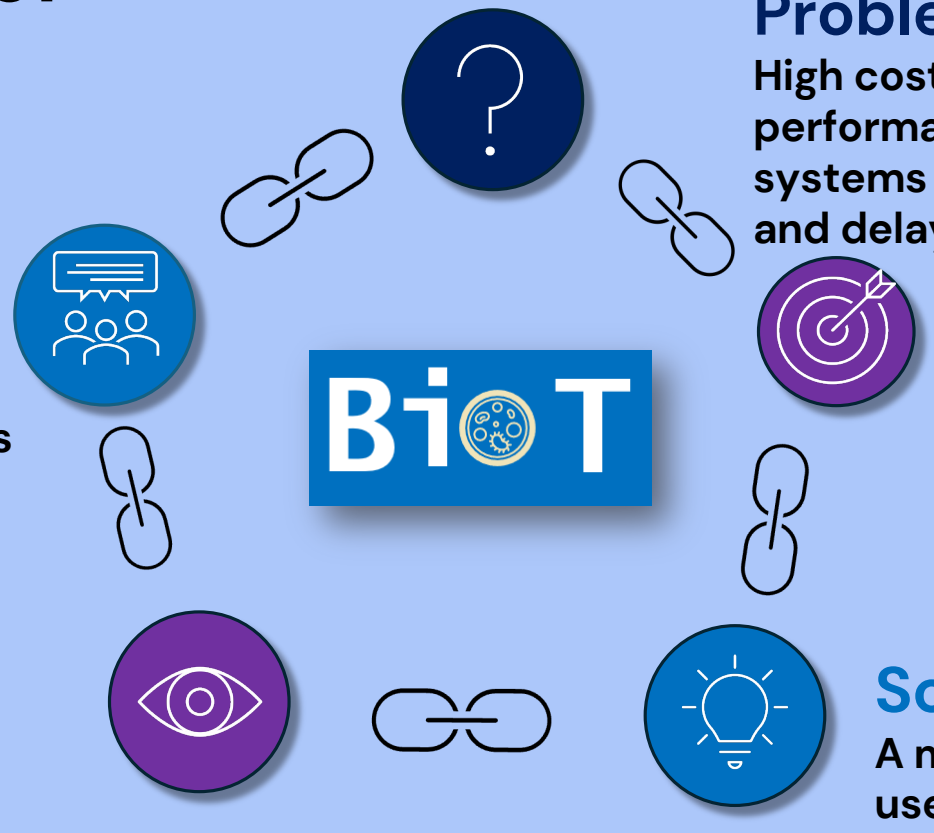
A modular, reliable and easy to use nutrient delivery system that acts as a platform for bioreactors

Vision

Revolutionise treatments, accelerate healing and reduce costs

Target Audience

Biotech firms, clinics and patients



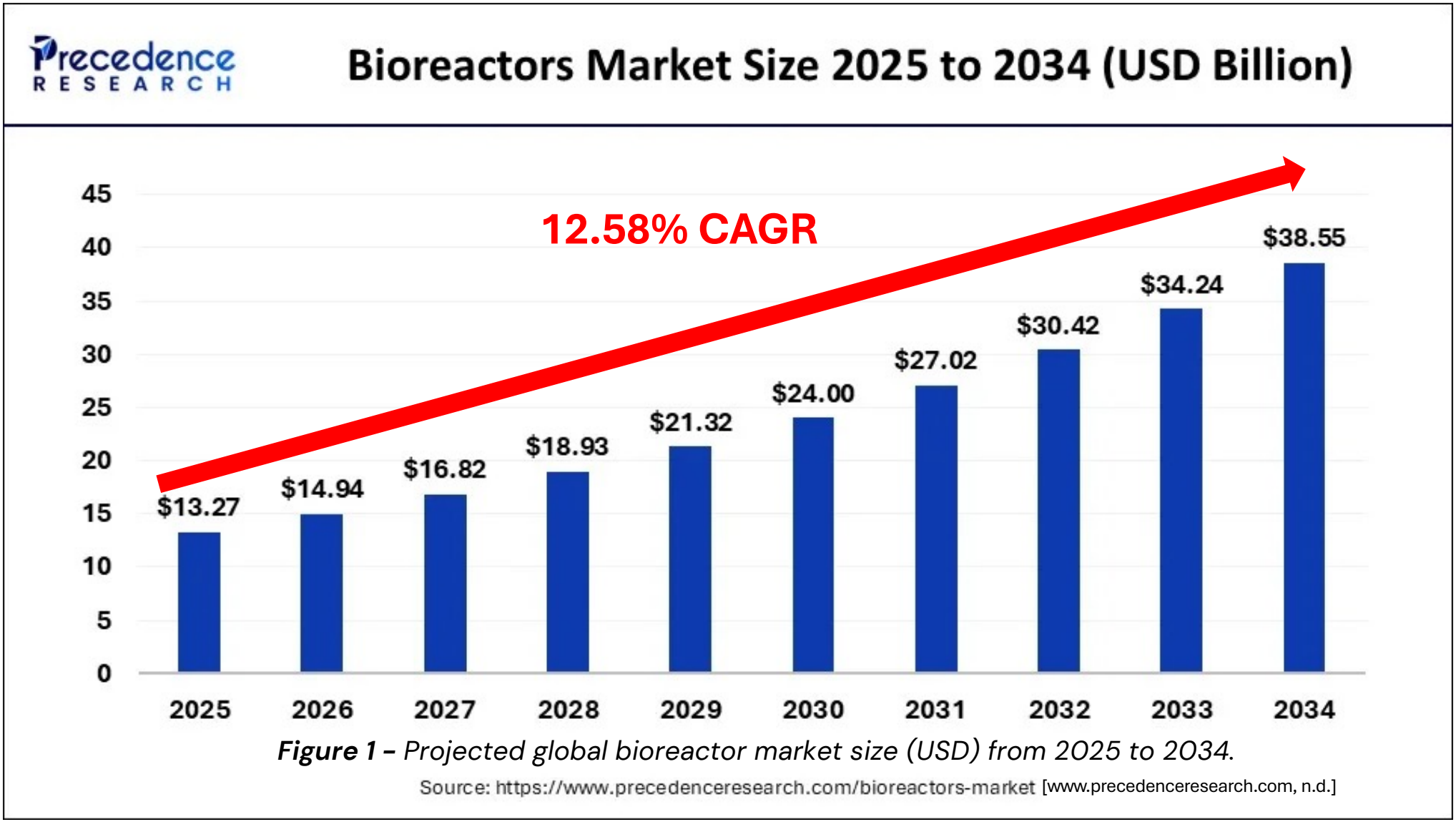


Figure 1 – *Projected global bioreactor market size (USD) from 2025 to 2034.*

Source: https://www.precedenceresearch.com/bioreactors-market [www.precedenceresearch.com, n.d.]

Organ Damage

- Organ damage may happen due to degenerative diseases, congenital diseases or injuries
- Organs have limited capacity for self-healing
- Complete restoration is often deemed impossible

Current Methods of Tackling Organ Loss:



Implantable medical devices



Organ and tissue transplantations

Medical implants



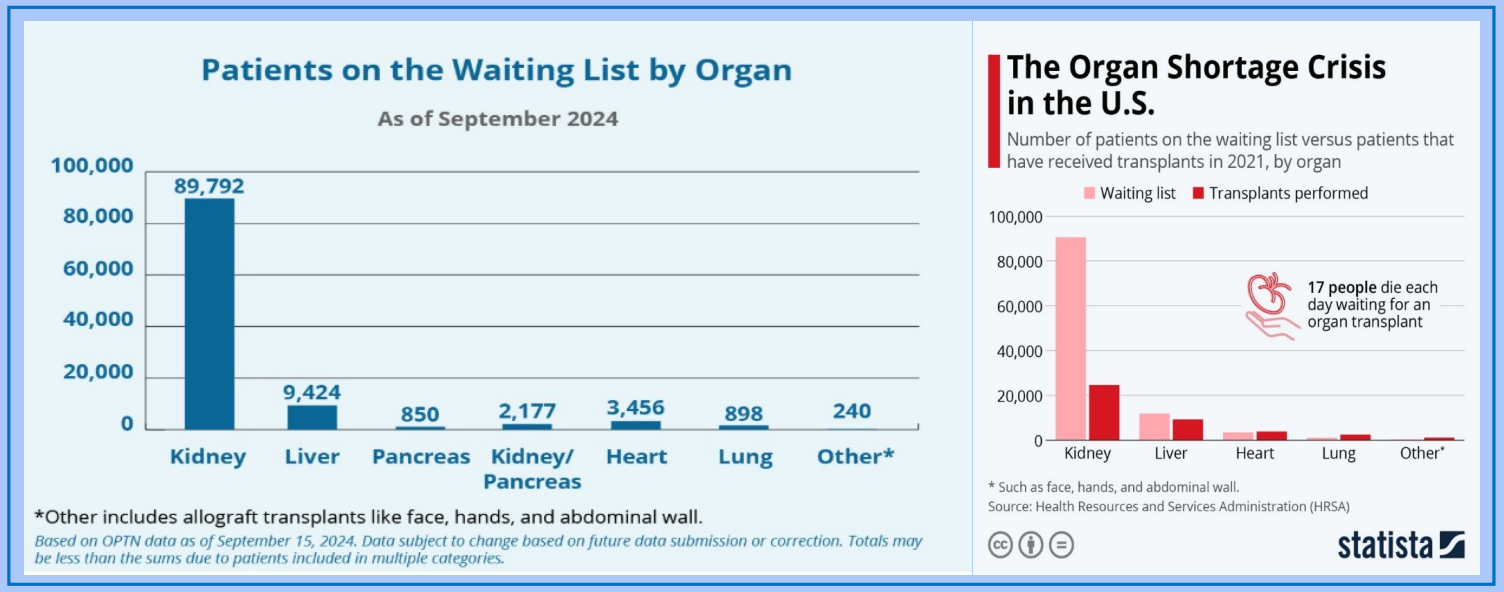
- Limited lifespan
- Repeated surgical intervention required to replace them
- Adds cost and time
- Affects patient's wellbeing
- Not ideal for active age groups

Organ Transplantations

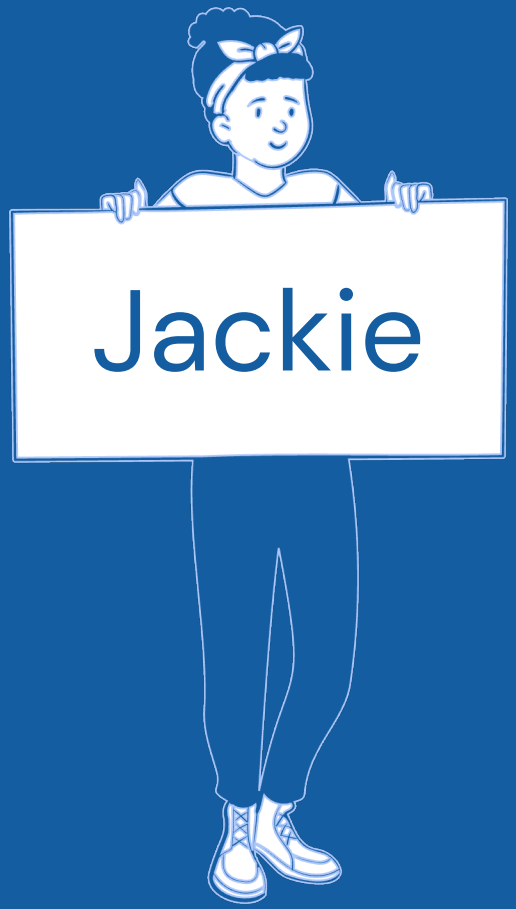
17 people die each day waiting for an organ transplant

Risks of transplant rejection and chronic pain

Ethical complications



Sources: (Fleck, 2022; HRSA, 2025)



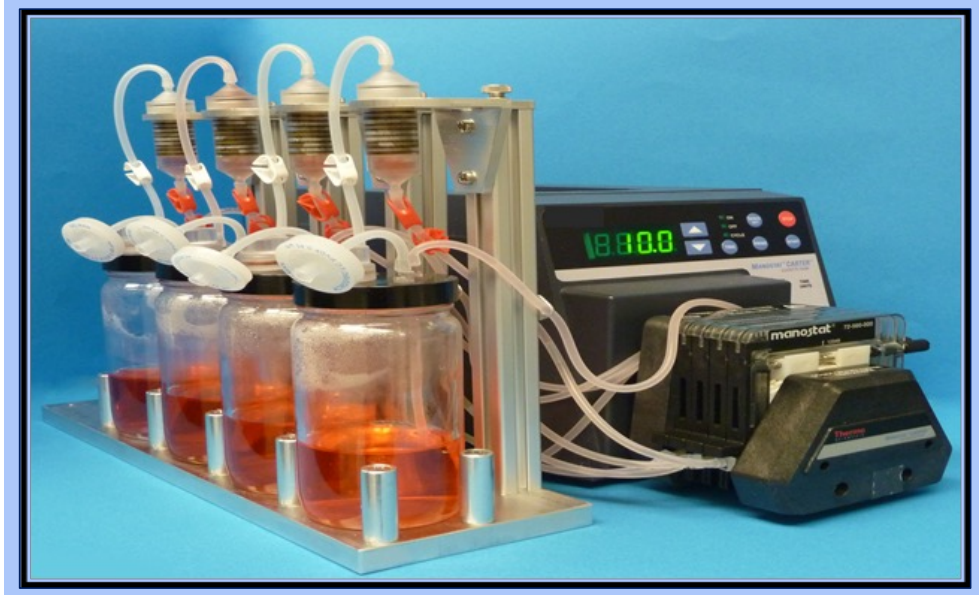
The Demand for Bone Tissue Engineering

- Cells cultures are unpredictable and highly sensitive to their surroundings
- 3 million+ people in the UK are affected by osteoporosis
- Fragility fractures are a consequence of this condition
- 500,000+ people in the UK need hospital treatment every year for fragility fractures

Source: (NHS Inform, 2024)

Perfusion Bioreactors

- ❑ Devices that maintain stable conditions to grow tissues
- ❑ Use mesenchymal stem cells



Advantages	Drawbacks
Can create 3D tissues	Highly sensitive, which causes uncontrolled cell death
Precise control of environmental conditions	Prone to human error
Continuous supply of nutrients, matches *in-vivo* conditions and supports tissue growth	Costly
Simple to operate	Can't be adopted for widespread clinical use

Team-based Innovation Strategy



1 Team Operating Agreement

2 Generating Ideas

3 Radical Candour

4 Financial Literacy

5 Solution proposal

Research

"Tissue engineering focuses at developing methods and technologies to synthesise in-vitro tissue constructs, which closely imitate the morphological, biochemical and mechanical features of in-vivo tissues " [Hansmann et al., 2012]

"Tissue engineering supports the creation of organ models for studying the effects of pathological conditions and testing the effect of therapeutics" [Hansmann et al., 2012]

"Perfusion bioreactors provide more uniform mixing of the media hence allowing for a better environmental control and physical stimulation of the cells in large constructs." [Gaspar, Gomide and Monteiro, 2012]

"A critical barrier to the development of 3D tissues in vitro is the limited transport of nutrients and metabolites within the construct interior" [Cartmell et al., 2003]

Effects of Medium Perfusion Rate on Cell-Seeded Three-Dimensional Bone Constructs *in Vitro**

SARAH H. CARTMELL, Ph.D., BLAISE D. PORTER, B.A., ANDRÉS J. GARCÍA, Ph.D., and ROBERT E. GULDBERG, Ph.D.

- **Scaffold preparation:** Freeze-dried, γ irradiated human trabecular bone rehydrated, cut and drilled into hollow cylindrical scaffolds (0.25 in diam/length)
- **Cell Seeding and Culture:** Scaffolds were seeded with 2 million MC3T3-E1 osteoblast cells and cultured in custom 3D perfusion bioreactor at various flow rates (1.0-0.01 mL/min) or under static conditions, for up to 7 days
- **Analysis techniques:** Cell proliferation was measured via DNA quantification

Key takeaway: Modelling and maintaining nutrient flow rate improves tissue regeneration

Results

- Highest cell proliferation at 0.01 mL/min
- Considerably higher for perfusion compared to static conditions

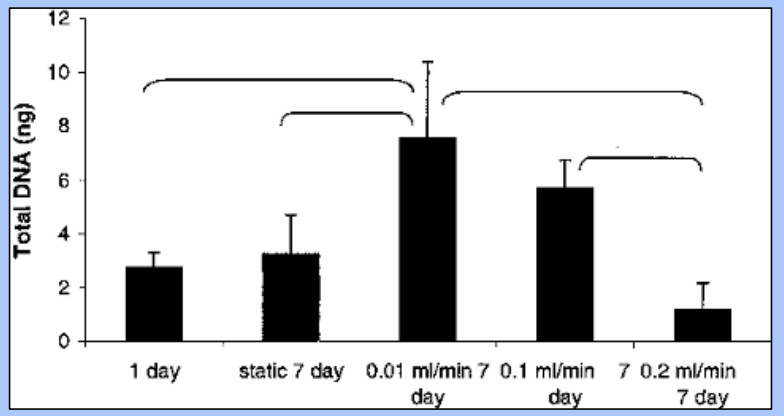


Figure 2 – *DNA assay after 1 week of perfusion*

Source: (Cartmell et al., 2003)

Design and Validation Process

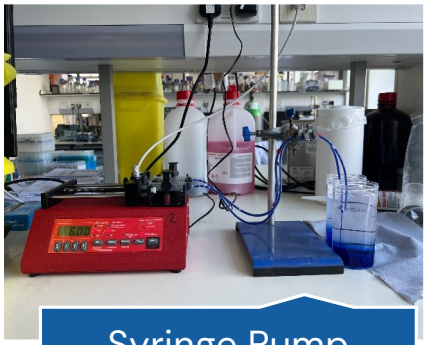
☑ = Complete



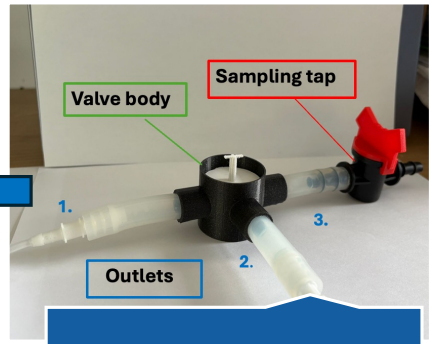
Peristaltic Pump Evaluation ☑



Holder Design ☑



Syringe Pump Evaluation ☑



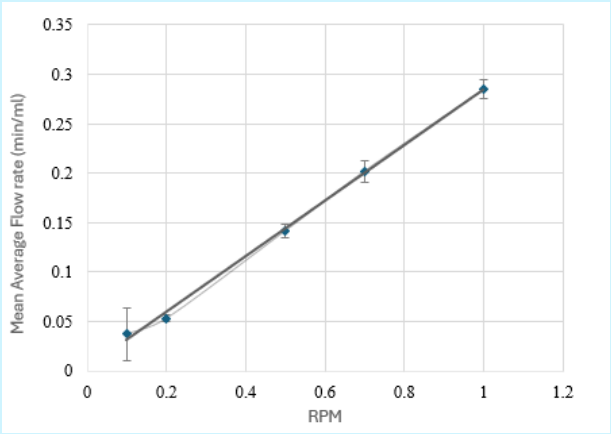
Valve System ☑



Flow Sensors ☑

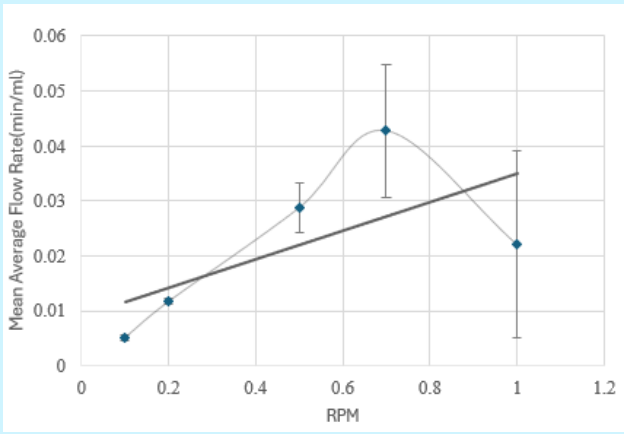
FINDING MOST ACCURATE TUBE SIZE AND OPTIMAL RPM

Research conducted by Jasmin Köhnken-Sawall found that the flow which induced the most osteogenic proliferation was 0.167min/ml (Köhnken–Sawall, 2011).



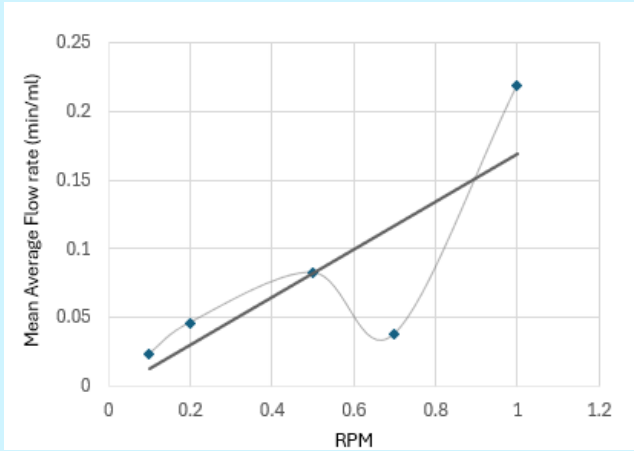
Tube 1 :
Inner Diameter: 2.0mm
Outer Diameter:4.0mm
Thickness:1.0mm

[a]



Tube 2:
Inner Diameter: 1.0mm
Outer Diameter:3.0mm
Thickness:1.0mm

[b]



Tube 3:
Inner Diameter: 2.4mm
Outer Diameter:4.0mm
Thickness:0.8mm

[c]

Figure 3 – *Graphs displaying mean average flow rates and standard deviations of the different tubes (a) Tube 1, (b) Tube 2 and (c) Tube 3 across tested RPMs*

What is the best tube configuration and RPM?

Chosen tube:

- Inner Diameter: 2.0 mm
- Outer Diameter: 4.0 mm
- Thickness: 1.0 mm

- This tube size demonstrated the lowest standard deviation across the tested RPM range, as well as the highest R-squared value compared to the other tubes.

- At the tested RPMs, a flow rate of 0.167 mL/min was achieved, indicating that this value is not a prediction but an experimentally confirmed result.

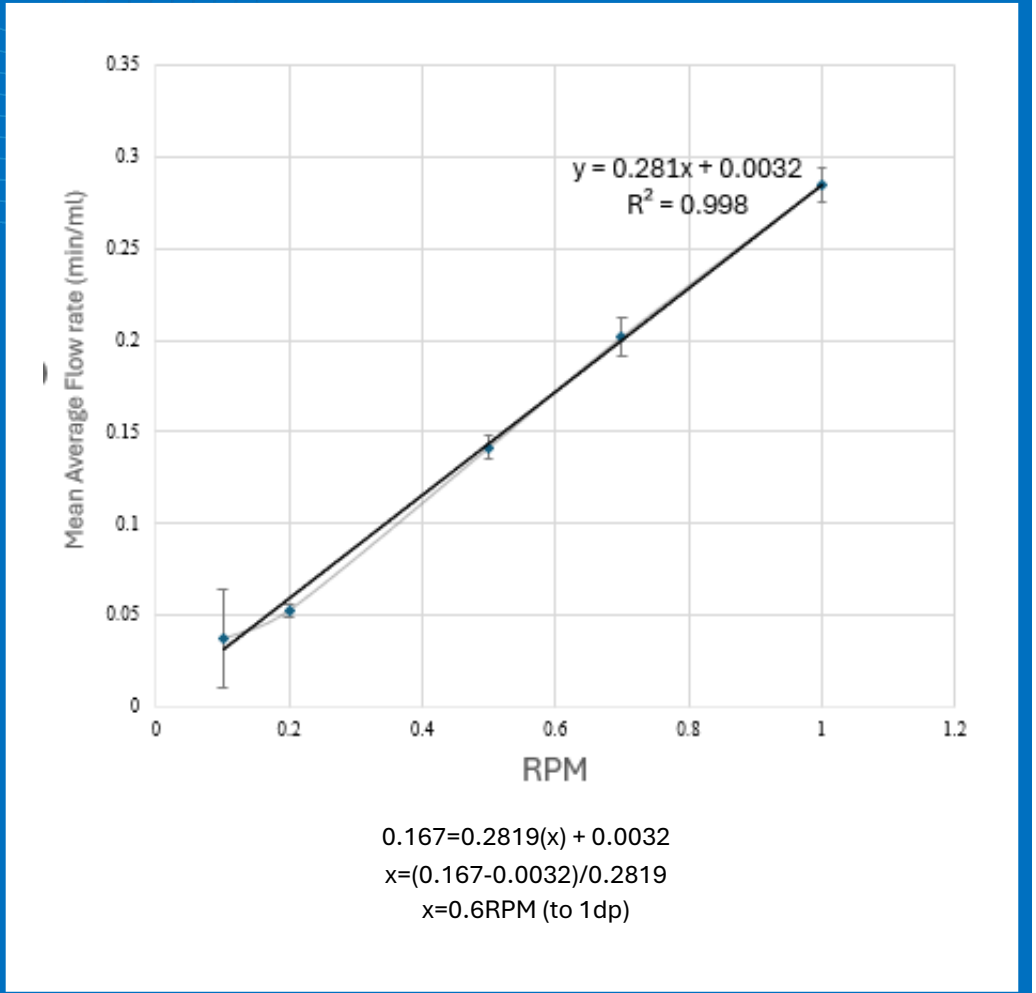
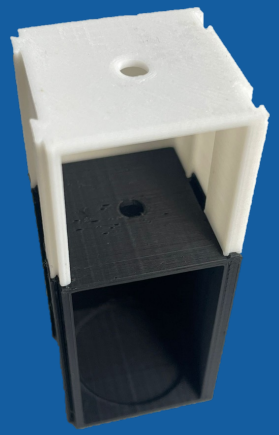


Figure 4 – *Graph of Tube 1 displaying mean average flow rates and standard deviations against tested RPMs*

Bioreactor Reservoir and Chamber Holders





Straightforward organisation

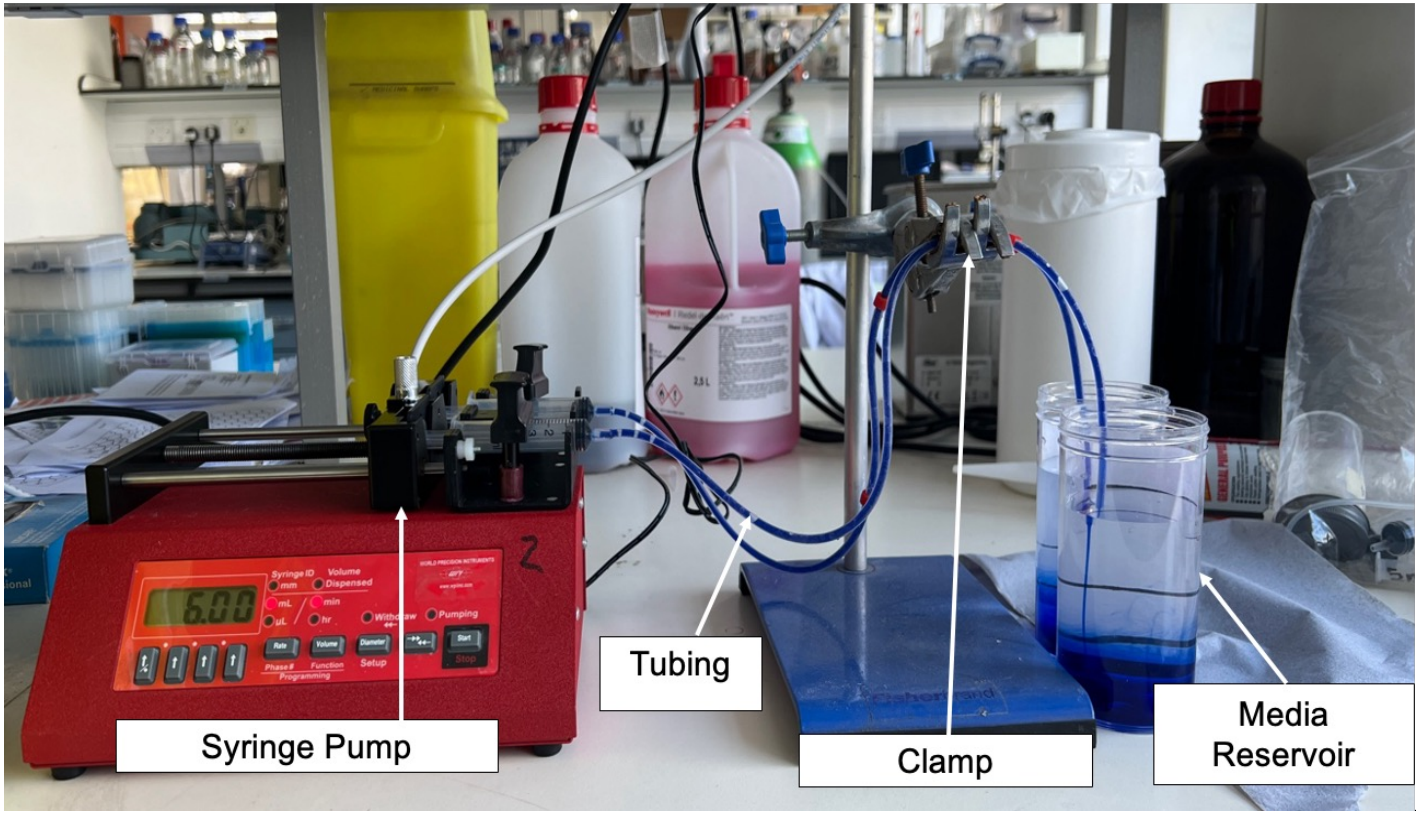
Modular

3D printed

Designed for efficient shipping and easy assembly

Sustainable materials – PLA and CPE

Syringe Pump for Nutrient Delivery



- Accurate
- Sterile
- Repeatable

Experimental Evaluation

Poor vertical mixing

- At low RPMs (2-4)
- Significant differences between top & bottom

Highest variability at bottom position

- At RPM 6

Moderate mixing

- At high RPMs (8-10)
- Bottom position is inconsistent

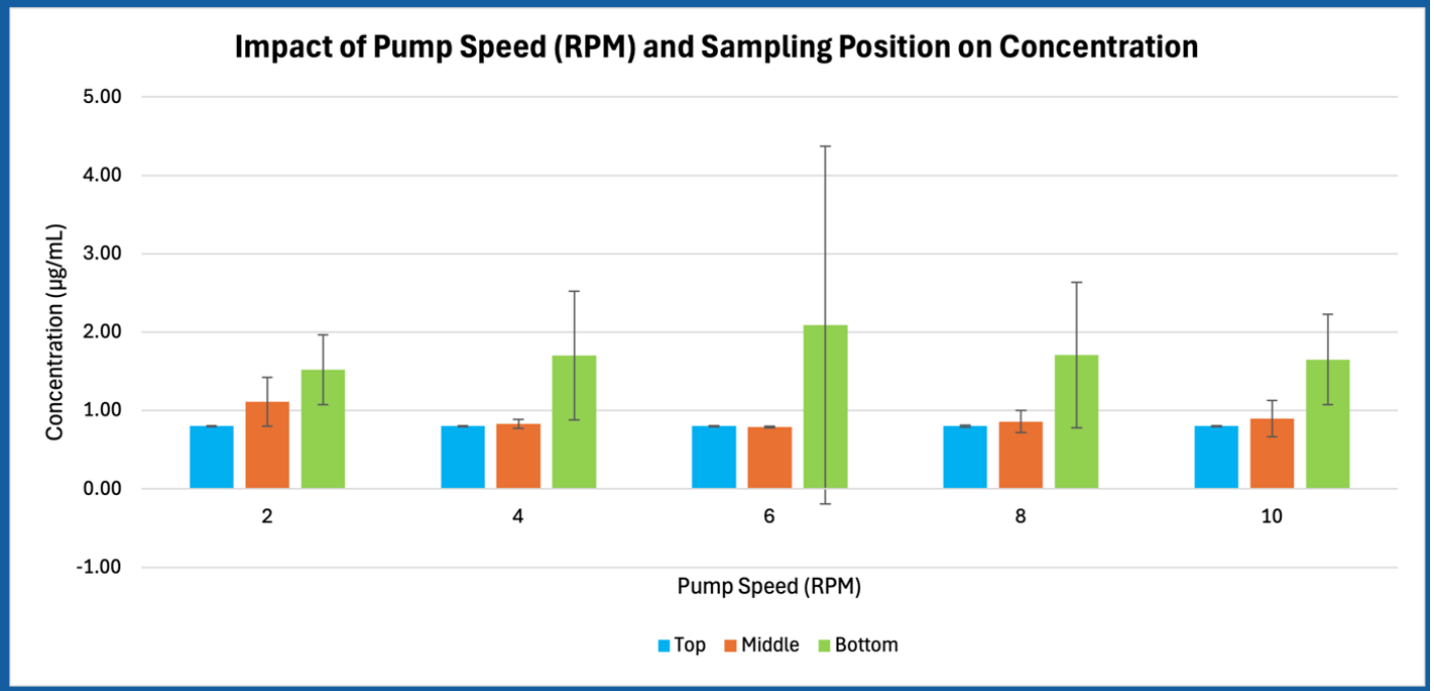
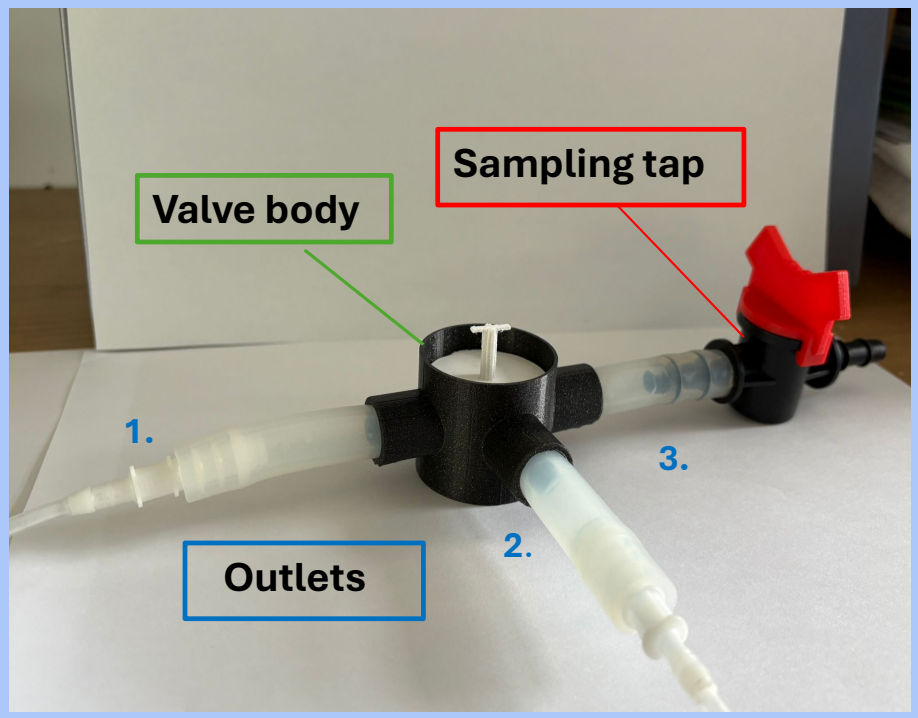


Figure 5– *Concentration measured at top, middle and bottom sampling positions across pump speeds ranging from 2 to 10 RPM.*

Valve-tap Sampling System

- 3D printed components, cost effective
- Fits in-line with the rest of the system
- Air-tight sealing, minimises leakage
- Biocompatible and sustainable materials ensuring secure sampling
- Highly modular- all components can be easily removed



Experimental Testing
Model 1

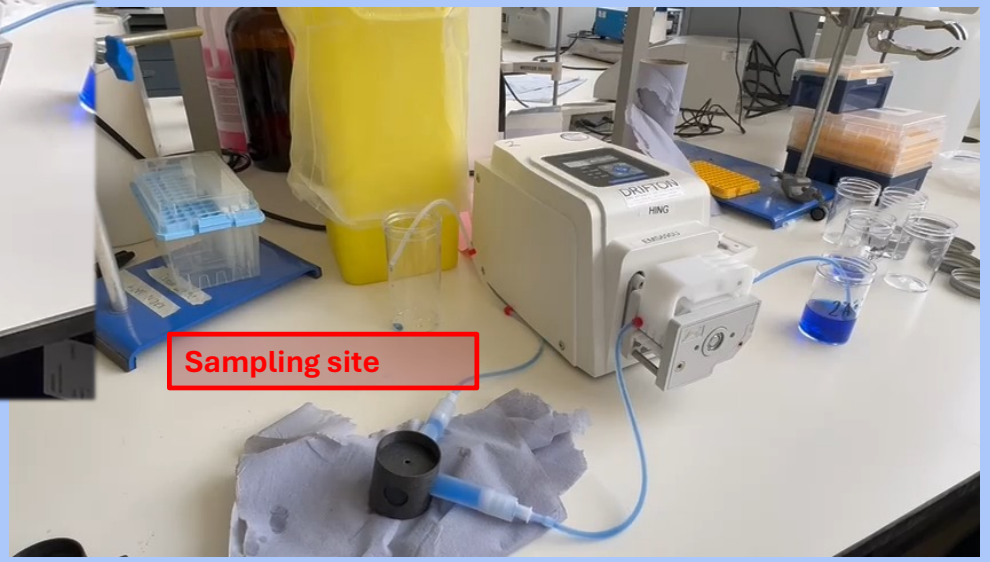
- Permits fluid flow



- Non-reliable

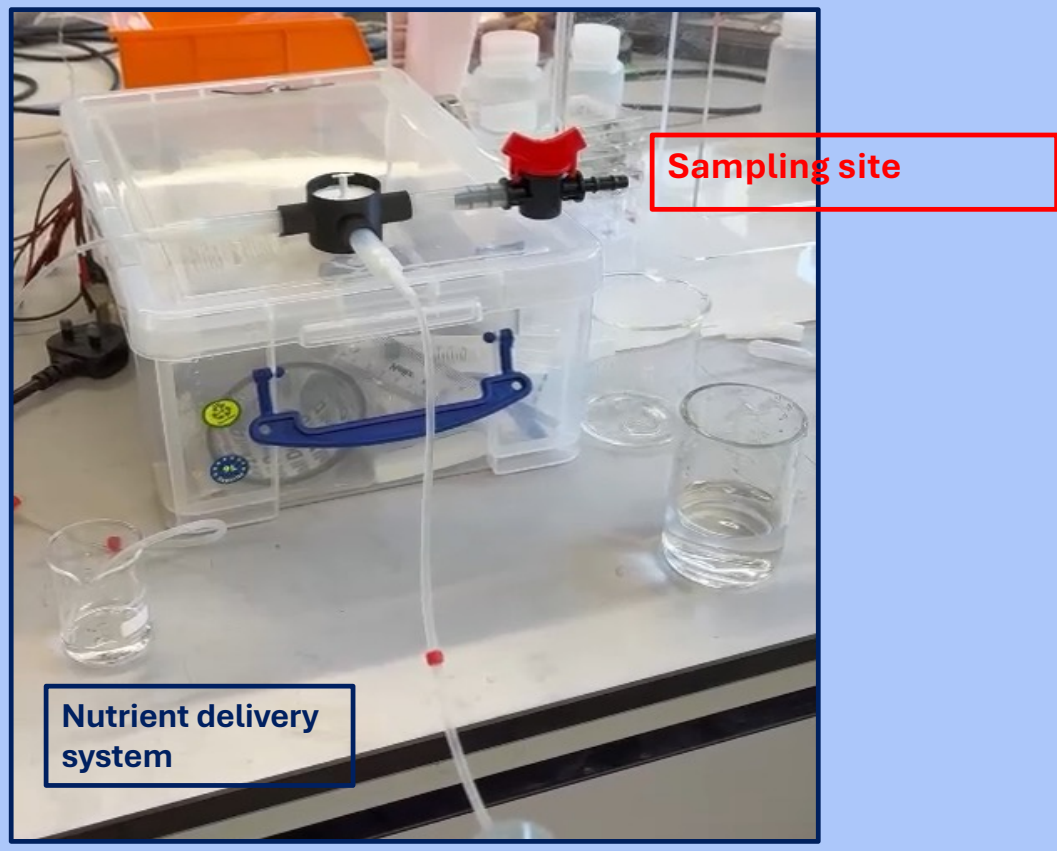
- Non-continuous process

System must be reset prior to taking a sample



Experimental Testing
Model 2–Sampling tap integration

- Safe extraction of samples
- Continuous operation
- In line-sampling



FLOW SENSOR INTEGRATION

Why is it important?

To enable real time monitoring and control of flow conditions occurring in the cell culture.

Optimisation Experiments

- Experiment 1: Compared **2.0 mm vs 2.4 mm tubing**
- Experiment 2: Tested **occlusion levels** (3–9) for optimal flow regulation
- Experiment 3: Investigated **external vibration** and damping using acoustic foam

Sensor Configuration settings:
Resolution: **11-bit**
Aquisition Frequency: **10 Hz**
Damping: acoustic foam (6 mats)

Sensor: Elveflow BFS



Optimal Configuration for Accurate Readings

Tubing: 2.4 mm ID
➤ Reduced noise & flow variability

Vibration Control: Acoustic foam
➤ Improved Signal-to-Noise Ratio



Pump Occlusion Level: 7
➤ Most stable pulsatile flow

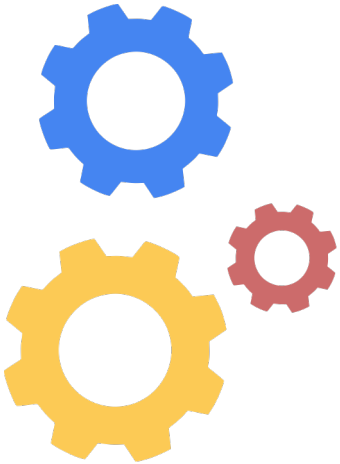
Sensor Placement: Downstream of scaffold
➤ Reflects actual flow experienced by cells



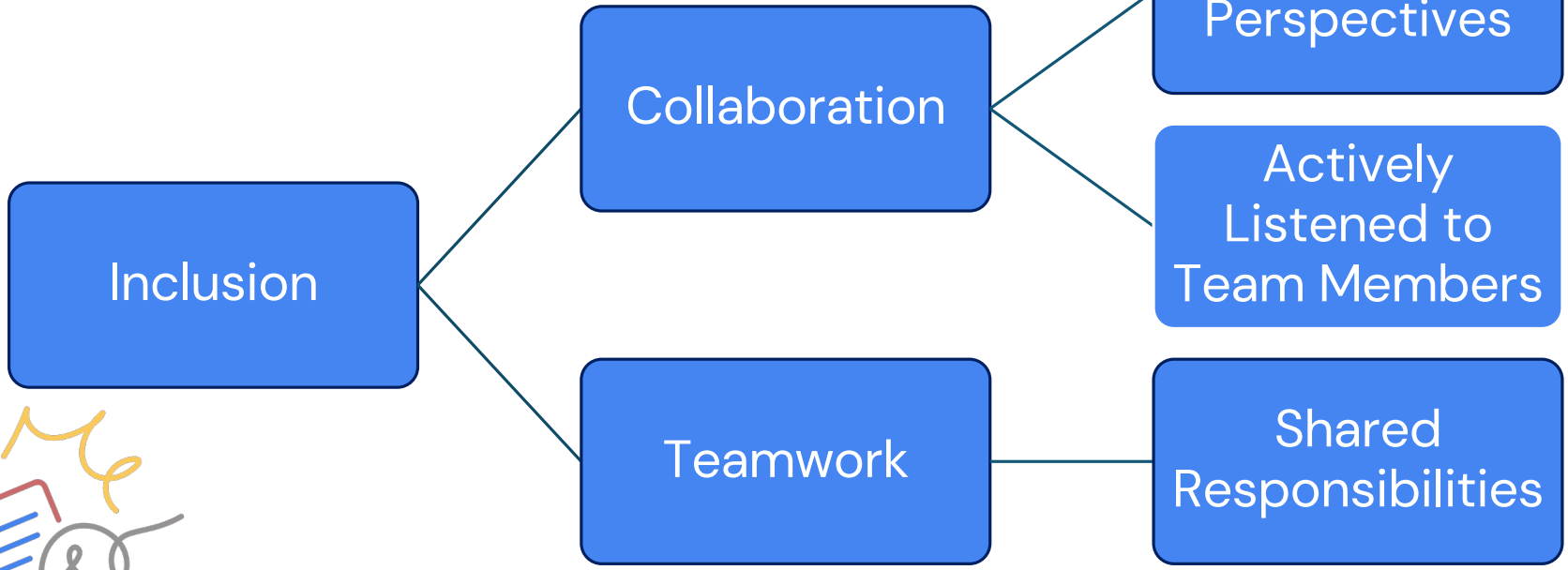
THE BENEFITS OF OUR SYSTEM

- ❑ Reduce Animal Testing
- ❑ Reduction of Waste Material
- ❑ Reliable Treatment Option
- ❑ Marketability
- ❑ Reduced likelihood of Rejection

Reflection and Next Steps

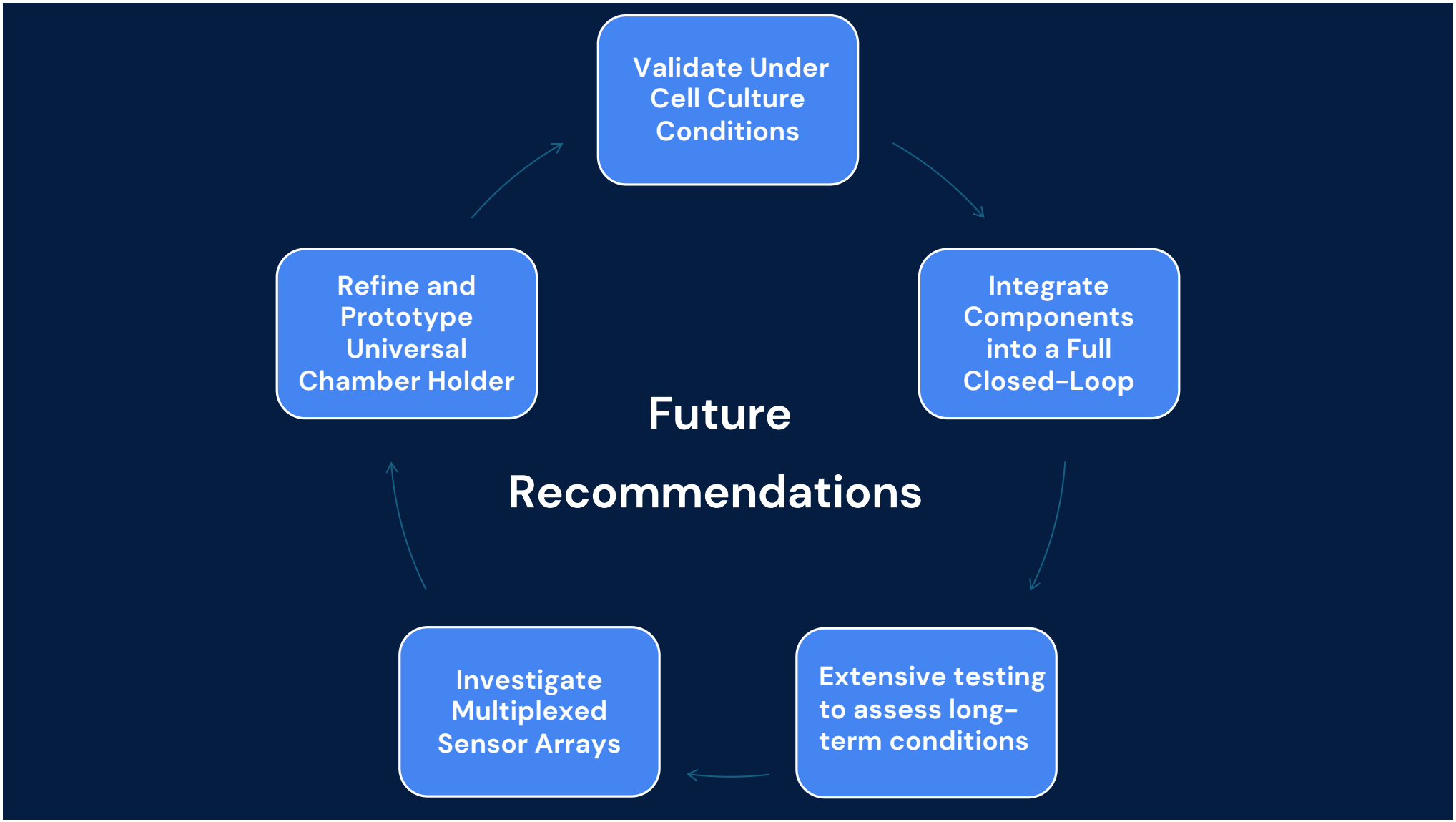




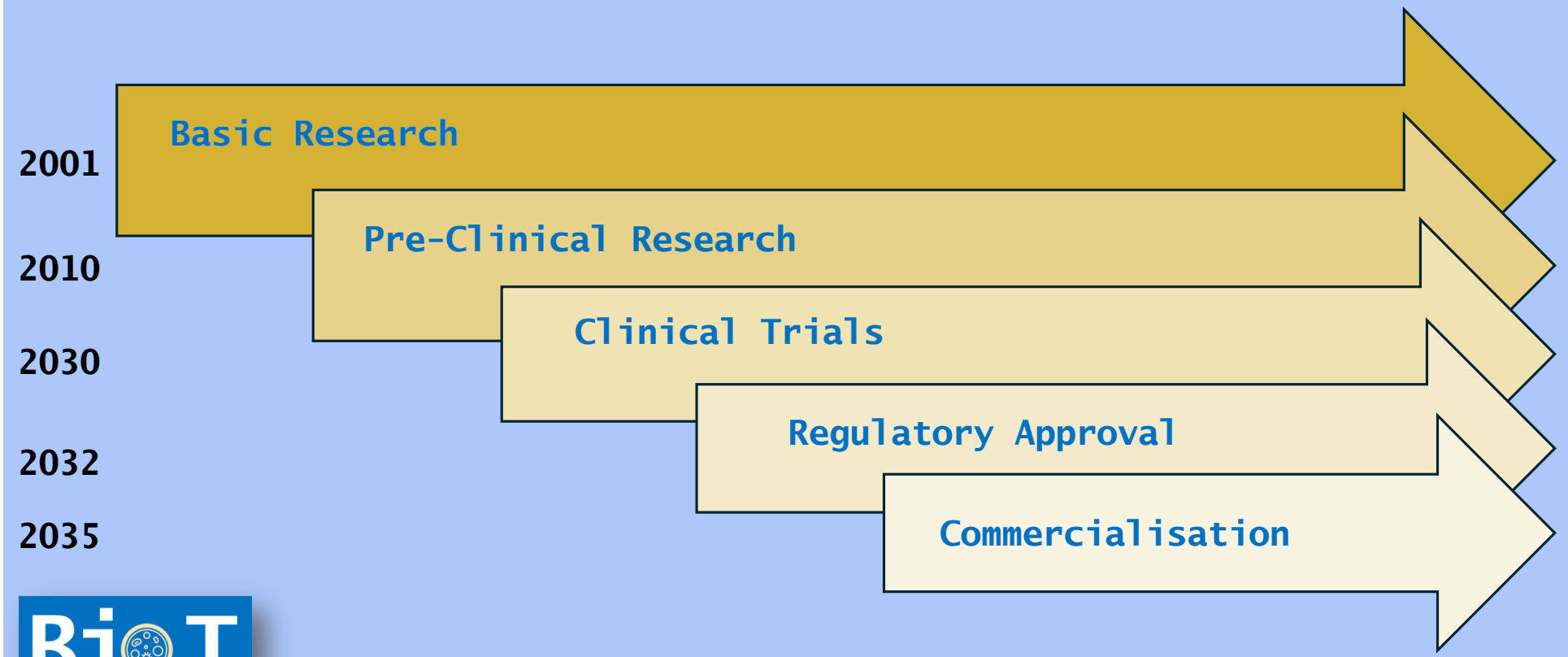


Risk Evaluation and Mitigation

Risk type	Description	Mitigation Measures	Likelihood	Severity
Operational	Air bubbles disrupting nutrient flow	Tubes submerged; optimal tubing selected	2	3
	Flow fluctuations due to vibration	Acoustic foam damping	2	2
	Syringe pump flow inconsistency	Calibrated RPMs; midpoint mixing tested	2	3
	Software/sensor instability	ESI reconfigured; data backups after runs	2	2
Health and safety	Electric shock from fluid contact with powered equipment	Insulated cables; elevated setup; dry area	1	4
	Burns/fume exposure from 3D printing	Low-temp PLA; ventilation; PPE	2	2
	Pinch hazard from pump rollers	PPE; tied hair; supervised operation	1	3
	Spills or splashes during fluid handling	PPE; dry bench; proper protocols	2	2


Validate Under Cell Culture Conditions
Integrate Components into a Full Closed-Loop
Extensive testing to assess long-term conditions
Investigate Multiplexed Sensor Arrays
Refine and Prototype Universal Chamber Holder
Future Recommendations

Progress


2001
Basic Research
2010
Pre-Clinical Research
2030
Clinical Trials
2032
Regulatory Approval
2035
Commercialisation

Executive Summary

BioT

Problem

High costs and inconsistent performance of current bioreactor systems hinder effective treatment and delay large-scale adoption

Mission

Improve reliability, adaptability and scalability of bioreactors.

Motivation

- Improve patient treatment, reduce reliance on animal testing and support clinical adoption.

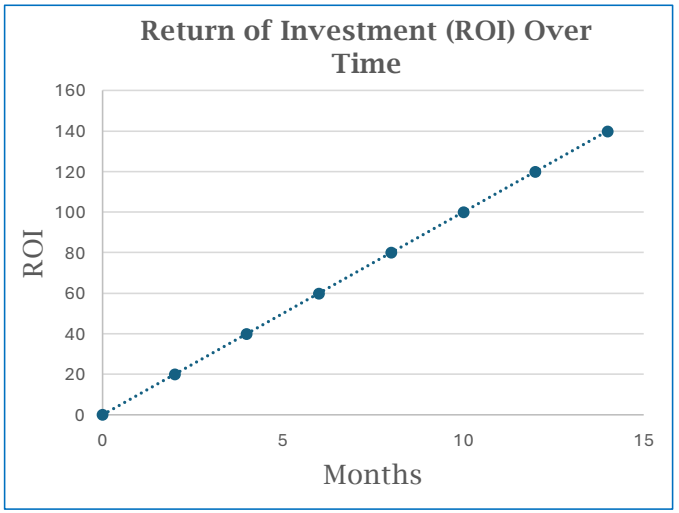
Target

- Undergo regulatory approvals and commercialisation by 2035.

Impact

- Create a reliable system with low variability, minimal waste and high cost-effectiveness.

Finance



Next steps

- Integrate closed-loop control, conduct extended-duration testing and validate with cell culture media.

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